

Q1. Find smallest and largest number

CODE

```
arr=[12,5,18,3,25,9]
smallest=arr[0]
largest=arr[0]
for num in arr:
    if num<smallest:
        smallest=num
    if num>largest:
        largest=num
print("smallest number:",smallest)
print("largest number:",largest)
```

OUTPUT

(no output)

Q2. Missing Number

CODE

```
nums=list(map(int,input().split()))
n=len(nums)+1
expected_sum=n*(n+1)//2
actual_sum=sum(nums)
print(expected_sum-actual_sum)
```

INPUT

4 2 1 6 5

OUTPUT

(no output)

Q3. Second largest number

CODE

```
nums=list(map(int,input().split()))
unique_nums=list(set(nums))
unique_nums.sort()
print(unique_nums[-2])
```

INPUT

12 5 18 3 25 9

OUTPUT

(no output)

Q4. Most repeated number

CODE

```
nums=list(map(int,input().split()))
max_count=0
most_repeated=nums[0]
for i in nums:
    count=nums.count(i)
    if count>max_count:
        max_count=count
        most_repeated=i
print(most_repeated)
```

INPUT

2 4 6 2 5 3 1 2

OUTPUT

(no output)

Q5. Unique elements

CODE

```
nums=list(map(int,input().split()))
unique_elements=[]
for i in nums:
    if nums.count(i)==1:
        unique_elements.append(i)
for x in unique_elements:
    print(x,end=" ")
```

OUTPUT

(no output)

Q6. Rotate an array by 'N' positions

CODE

```
arr=list(map(int,input().split()))
rotated=arr[2:]+arr[:2]
for x in rotated:
    print(x,end=" ")
```

OUTPUT

(no output)

Q7. Remove all duplicates in an array

CODE

```
nums=list(map(int,input().split()))
result=[]
for i in nums:
    if i not in result:
        result.append(i)
for x in result:
    print(x,end=" ")
```

INPUT

2 4 6 2 7 8 3 4

OUTPUT

(no output)

Q8. Print all 0's followed by non-zero without changing the order

CODE

```
nums=list(map(int,input().split()))
zeroes=[]
non_zeroes=[]
for x in nums:
    if x==0:
        zeroes.append(x)
    else:
        non_zeroes.append(x)
result=zeroes+non_zeroes
for x in result:
    print(x,end=" ")
```

INPUT

1 0 4 0 2 5 0 7 0 6 0

OUTPUT

(no output)

Q9. count vowels, consonants, digits, and spaces

CODE

```
s=input()
vowels=0
digits=0
consonants=0
spaces=0
for char in s:
    if char.lower() in "aeiou":
        vowels+=1
    elif char.isalpha():
        consonants+=1
    elif char.isdigit():
        digits+=1
    elif char.isspace():
        spaces+=1
print("Vowels:",vowels)
print("consonants: ",consonants)
print("Digits: ",digits)
print("spaces: ",spaces)
```

INPUT

yazhini24

OUTPUT

(no output)

Q10. Palindrome

Score: 10.00 TC: 2/2 passed

CODE

```
s=input()
if s==s[::-1]:
    print(s,"is a palindrome")
else:
    print(s,"is not a palindrome")
```

INPUT

saveetha

OUTPUT

saveetha is not a palindrome